

Which Aesthetic Criteria Hold a Hand-Drawn Diagram, Tested One at a Time Against a Tool’s Layout of the Same Graph

Vasili Gavrilov*

Draft, August 2026. Numbers marked^P are from the climber pilot of 2026-08-21 (40 to 60 diagrams per corpus, one seed) and are replaced when the descent is rerun on the library; the directional test’s numbers are from the whole 15 to 40 node band.

Abstract

Automatic layout tools place the boxes of a diagram by minimising a weighted sum of aesthetic criteria: few edge crossings, no overlapping boxes, edges of similar length. If people drew the same way, a diagram laid out by hand would be a local minimum of that sum and a tool started from it would find nothing to change. We test each criterion separately on 700 diagrams of 15 to 40 boxes whose coordinates a person chose, from 1,571 parsed: 305 WikiPathways biological pathways, 248 Reactome SBGN maps and 147 BPMN process models, three communities with different authoring tools. The test is direct: move each box 2% of the drawing width in each of sixteen directions and ask whether any move lowers the criterion; the same test is run on layouts of the same graphs by three tools, **neato**, **neato** with overlap removal, and **dot**. Under overlap every hand-placed box is held and the term is zero; so is every box in a layout by a tool that removes overlaps. Under uniform edge length and under stress no hand-placed box is held in any corpus, while **neato**’s layouts are held under stress at 1.00, so the test has power. The hand layout is nowhere near a minimum of any criterion that prices distance, and the three-term energy tools use, which spends 94% of its value on edge length, holds none of it; fitting the weights by inverse optimisation returns weight zero for edge length on every corpus for that reason. Alignment of boxes into shared rows and columns holds 0.52, 0.21 and 0.82 of hand-placed boxes, against 0.06 in **neato**’s layouts and 1.00 in **dot**’s; by HOLA’s gridiness 58%, 36% and 77% of hand-placed boxes are in an alignment of three or more, none of **neato**’s and 94%, 94% and 89% of **dot**’s. On every criterion but alignment the hand layout’s profile is the layered tool’s; the person aligns part of the drawing where **dot** aligns all of it. Crossings hold every box in every layout at this radius, because no single move of 2% removes one; the diagrams people draw are near-trees (median edges per box 1.06, 0.96 and 1.03, against 1.30 for the Rome benchmark) and WikiPathways authors leave about one crossing per 26 edges where **neato** leaves none. The measurement is deterministic and runs on a small C library for stochastic search whose descents and matched-budget random control supply the secondary estimand.

1 Introduction

A diagram is a graph with a drawing: boxes for the nodes, lines for the edges, and a position for every box. When a program chooses the positions it minimises an *energy*, a weighted sum of quantities the field has agreed are bad: edges that cross, boxes that overlap, edges of very unequal length, boxes that sit on edges they do not belong to [3, 2, 4, 5]. The weights are asserted by the tool’s author. The premise is that a person drawing by hand is doing the same thing more slowly, so that the drawing they accept is one a tool would also accept.

*Independent researcher. Code, parsed corpora and scripts: <https://github.com/Anode1/cjitter>, directory [example/diagrams](#).

The premise has a direct test, one criterion at a time. Take a diagram a person drew. Move one box a small distance in each of several directions and ask whether any of the moves lowers the criterion. If none does, the box is *held*: the person has put it where the criterion wants it, or where the criterion has stopped caring. Do this for every box and the fraction held says how much of the layout is a local minimum of that criterion; the criterion’s value says how well it is met. A criterion that holds every box at value zero is one the person satisfied; one that holds no box is one the person did not consider, or traded away.

We ran the test on 700 diagrams of 15 to 40 boxes from three unrelated communities (Section 4), and on layouts of the same graphs by three tools. The findings:

1. Under overlap every hand-placed box is held and the term is zero in every corpus. Every box in a layout by a tool that removes overlaps is held too; **neato** without overlap removal holds 0.39 to 0.76. Overlap is a criterion people satisfy exactly, in common with every tool that tries.
2. Under uniform edge length and under stress no hand-placed box is held: 0.00 in every corpus under either. The test has power: **neato**’s stress layouts are held under stress at 1.00 in every corpus. Under the three base criteria summed with the usual equal weights the length term dominates and no box is held, which is what an earlier version of this study, using a capped descent, reported as “hand layouts are not stationary”. Fitting the weights by inverse optimisation gives uniform edge length the weight 0.00 on every corpus for the same reason: any positive weight on a criterion that holds no box makes the sum hold no box.
3. Alignment of boxes into shared rows and columns, HOLA’s gridiness, puts 58%, 36% and 77% of hand-placed boxes in an alignment of three or more; **neato** puts none, **dot** 94%, 94% and 89%. The continuous form A1 holds 0.52, 0.21 and 0.82 of hand-placed boxes against 0.06 for **neato** and 1.00 for **dot**. Orthogonality of edges holds 0.16, 0.05 and 0.41, more than any tool; separation of boxes from foreign edges is nearly satisfied everywhere (value 0.006 to 0.022) and holds 0.76 to 0.86, which is the flatness of a term with little to act on. In WikiPathways a third of boxes share an exact coordinate with another box while 2% of coordinates are integers, so the alignment there is the person’s act; in BPMN it is partly the grid and mostly the lanes.
4. On every criterion but alignment the hand layout’s profile is **dot**’s: held under overlap, free under length and stress, partly held under orthogonality. The person draws like a layered tool that aligns part of the drawing.
5. Crossings hold every box in every layout at radius 0.02, because no single move of that length removes one, so the count is flat. The value separates: WikiPathways authors leave one crossing per 26 edges (0.038 of m), **neato** none, **dot** one per 8. The diagrams people draw are near-trees, median edges per box 1.06, 0.96 and 1.03 against 1.30 in the Rome benchmark [16]; crossings are what the theory is about [1], and on these diagrams there are few to remove.

The rest of the paper is the foundation behind the test (Section 2), the instrument (Section 3), the corpora, the method, the measurements, the alternatives not yet excluded (Section 7), and what is already known (Section 8). A tool builder reading only this far has the changes: take the weight off edge length, add alignment, and print the fraction of boxes the energy holds when started from a layout a person accepted.

2 Foundations

The energy. A layout of a graph with N nodes is a point $x \in \mathbb{R}^{2N}$, two coordinates per node, the boxes' sizes held fixed. An aesthetic energy is

$$E_w(x) = \sum_{k=1}^K w_k T_k(x), \quad w_k \geq 0, \quad (1)$$

where each term T_k is a quantity to be made small. The tool's author asserts the w_k ; the field of multicriteria layout is the choice of the T_k and the w_k [2, 4].

Local minimum. A point x is a local minimum of E if there is a radius $r > 0$ with $E(y) \geq E(x)$ for every y within distance r of x . For a differentiable E this implies $\nabla E(x) = 0$. Two of our terms are not differentiable: crossings is a count, piecewise constant in x , and overlap has corners where boxes touch. So the test uses no gradient. Fix a radius d and a set V of unit directions in the plane (sixteen, equally spaced). For node i and direction v let $x + d v^{(i)}$ be the layout with node i moved by d along v and every other node fixed. Node i is *held* at radius d if

$$E(x + d v^{(i)}) \geq E(x) \quad \text{for every } v \in V, \quad (2)$$

and the estimand is

$$q_d(x) = \frac{1}{N} \#\{i : \text{node } i \text{ is held at radius } d\}. \quad (3)$$

$q_d = 1$ says the layout is a local minimum of E over single-node moves of length d ; for smooth E and small d it says $\nabla E(x) = 0$ to within the resolution of V . Because (2) is a comparison of two energies it is defined for every term, and because it is evaluated per term it separates the criteria a person satisfied from those they did not. A descent-based measure (how far a capped descent moves the layout, the estimand of the first version of this study) answers the same question for the sum and cannot separate the terms; it is retained as a secondary.

The inverse problem. Given layouts $x^{(1)}, \dots, x^{(G)}$ that people accepted, inverse optimisation asks for the weights w under which they are nearest to stationary [18, 19]. With the directional differences $D_{iv}T_k = T_k(x + d v^{(i)}) - T_k(x)$ in hand, node i is held under w exactly when $\sum_k w_k D_{iv}T_k \geq 0$ for all v , a set of linear inequalities in w . The residual is the smallest total violation over w on the simplex, a linear program, and it is zero only if some weighting of the terms makes the layout a local minimum. A term with $D_{iv}T_k < 0$ for some v at almost every node (uniform edge length, at any layout whose edges are not all the same length) cannot take positive weight without violating the inequalities at those nodes, which is the zero weight the random-search fit found.

Step terms. A term that is a count (crossings) or a step function of position (gridiness, Section 5) is flat almost everywhere, so it holds a node wherever no single move of length d changes it. For such a term q_d says how much of the layout the criterion has stopped acting on, which is a different thing from how well the criterion is met, and the term's value at the layout is reported beside q_d .

The reference length. The length and stress terms compare distances with a reference L . Fixed at the layout's median edge length, L is in general not the scale at which the term is least for that layout, so at a true minimum every node still has a scaling direction that lowers the term and the converged reference is not held (neato's stress layouts: $q_{0.02} = 0.07$ to 0.12). L is therefore the value at which the term is least for the layout as loaded, $\sum \ell^2 / \sum \ell$ over edges for length and the same over $\|x_i - x_j\| / \delta_{ij}$ for stress, fixed before any move; neato's stress layouts are then held at 1.00 in every corpus. The median and $1/\sqrt{N}$ are sensitivities.

The null and the controls. A test that can only succeed is not a test. The *stationarity reference*: q_d on a layout a descent has converged on must be near one, or the test lacks power for that energy; *neato*’s layouts are the reference for stress (1.00 in every corpus) and the capped descents of the pilot for the sums (0.94 to 1.00^P). The *tool controls*: q_d on layouts of the same graph with the same box sizes by **neato** (stress, boxes left where they overlap), **prism** (*neato* followed by its overlap removal) and **dot** (layered), which say what a layout that does minimise something scores on each term. The *matched-budget control*: uniform random placement at the same number of energy evaluations, which the hand layout must beat, or “held under overlap” would be a property of sparse graphs and not of the person.

3 The instrument

The descents, the control and the paired statistics come from **cjitter**, a C library of four stochastic searches over a box of real variables: hill climbing with restarts, simulated annealing, a genetic algorithm, and uniform sampling as the control. Every run is byte-reproducible from its seed on every platform, every method constant is a visible field, and **cjitter_compare** runs all four at the same budget on the same seeds and reports an exact paired sign test, Holm-corrected, against the control. “Better than uniform sampling at equal cost” is the only claim it lets a search make.

It is the latest of a series of small tools by the author, each reporting the condition under which its own answer is wrong:

year	tool	what it reports against itself
1997	backpropagation thesis [32]	descent on an error energy, derived
2001	label placer	per-label random steps, kept if the summed overlap fell
2011	least squares, now linearr	when a line is the wrong shape
2025	bpnn	when the network has memorised the rows
2026	cjitter	when a search does no better than sampling

The 1997 thesis derives backpropagation as gradient descent on an error surface; the 2001 placer, deployed in industry, is the same descent without a gradient, one label moved a random step and the step kept when the total cost fell. That operator, a proposal that moves one object and is judged on the summed energy, is the **block** field of the library, and the directional test of equation (2) is the same proposal made exhaustive: every direction tried, none accepted, the answer read off. The present paper starts from where a person placed the objects and reads which terms pull.

4 The corpora

Three collections of diagrams whose node coordinates a person chose, from three communities with different authoring tools, all redistributable.

corpus	licence	files	parsed	15–40 nodes	median m/n
WikiPathways GPML, <i>Homo sapiens</i>	CC0	1016	571	305	1.06
Reactome SBGN-ML	CC0	1382	701	248	0.96
BPMN Academic Initiative	academic	[PENDING: n]	299	147	1.03

The layouts are working documents: Reactome diagrams are laid out by curators [27], WikiPathways pathways are drawn in PathVisio [28], and the BPMN models are students’ work in Signavio [29, 30]. Nobody was asked to draw nicely. The three tools share no layout code.

Two WikiPathways files (WP5037, WP3391) carry coordinates converted from Reactome [31] and are dropped.

A diagram enters as its largest connected component (194 of the 248 Reactome graphs and 54 of the 147 BPMN graphs in the band are the component of a larger diagram); container glyphs (SBGN compartments, BPMN pools and lanes) are not nodes; SBGN process glyphs are nodes, so the Reactome graphs are bipartite and their m/n is the process graph's; each layout is rescaled so that its bounding box fits the unit square, box sizes scaled with it. The directional test runs on 15 to 40 nodes; the climber pilot used 10 to 400. Duplicate pathways (ortholog copies of one human pathway) are removed by model identity. **[PENDING: counts after deduplication; the 41 to 100 node sensitivity]**

How the coordinates were made matters for the alignment criterion. Of the raw coordinates, 2% are integers in WikiPathways, 69% in Reactome and 71% in BPMN (46% multiples of five); the fraction of boxes sharing an exact x with another box is 0.33, 0.06 and 0.47, an exact y 0.28, 0.12 and 0.73. WikiPathways authors align by hand on a continuous canvas; BPMN authors work on a grid inside lanes; Reactome's curation tool rounds and its authors align little.

5 The method

Terms. With n nodes, m edges, box i of width w_i and height h_i , L the reference length of Section 2, fitted to the layout under test:

term	definition	zero when
T_C crossings	proper crossings between segments of two routes with no shared endpoint, $/m$	planar
T_O overlap	overlap area of boxes / total box area	no overlap
T_L length uniformity	mean over edges of $((\ell - L)/L)^2$	all edges of length L
T_R orthogonality	mean over edges of the length-weighted mean over its segments of $\min(\Delta x , \Delta y)/(\Delta x + \Delta y)$	routes axis-aligned
T_A alignment	$1 -$ fraction of nodes in an alignment of ≥ 3 (tol. 0.005)	every node aligned
T_N node-edge	squared shortfall of node to foreign route distance below $(w + h)/4$	none within
T_F flow	mean over directed edges of the backward component along the reading direction u , over length	no edge runs back

The base energy is $T_C + T_O + T_L$ with weights $(1, 1, 1)$, the form of the simulated-annealing layout of Davidson and Harel [2] and of the multicriteria tradition since [24]. The field's other default is stress [25, 26], $\sum_{i < j} (\|x_i - x_j\| - L \delta_{ij})^2 / \delta_{ij}^2$ with δ_{ij} the graph distance, which force-directed tools and **neato** minimise and which Dwyer et al. [12] found human choices track; it is the second base energy. T_R , T_A and T_N are the candidates for what the base energy misses. The alignment definition above is HOLA's gridiness [8], the published form, a step function of position. Three continuous forms are the sensitivities: A1, per node the distance to the nearest other node along whichever axis is nearer, which has a corner at exact alignment; A2, rows and columns priced separately; A3, a Gaussian kernel of width $s = 0.02$, smooth. The corner is what holds a box: A1 holds 0.52 of WikiPathways boxes and A3 holds 0.20.

Routes. An edge is the polyline the file stores: the attachment point on the source box, the interior waypoints, the attachment point on the target box; where the file stores nothing it is the chord between the centres. Crossings, orthogonality and node-edge separation are evaluated on the route; length and stress on the centres. When a node moves its attachment points move with it and the waypoints stay. The attachment point matters: closed at the centres, 29% of BPMN routes with a waypoint are axis-aligned, closed at the stored attachments 86%. What each format stores, as a fraction of the edges in the band: **[PENDING: the route table from make_corpus.py]**. GPML's Elbow and Curved connectors store their endpoints only

and PathVisio computes the bends at draw time, so those routes read as chords; a quarter of WikiPathways edges end on an anchor of another interaction, where the person drew no segment to the node and the last segment is manufactured. The chords are the sensitivity.

Reading direction. T_F needs a direction u . It is one of the four axis directions, the one under which T_F is least for the layout as loaded, fixed before any move, the same rule as for L . Undirected edge kinds (GPML graphical lines, BPMN undirected associations) do not enter it.

Directional test. Radius $d \in \{0.005, 0.01, 0.02, 0.05\}$ of the drawing width, sixteen directions, every node, each term alone and the sums of the tables. Primary cell $d = 0.02$. The directions are exact to double precision (a series evaluated in the first quadrant and rotated), and a move that lowers the energy by less than 10^{-12} of its value is a tie; an earlier build with directions off by up to 4×10^{-6} counted the shift that error made as an improvement and read alignment lower by up to 0.04. Deterministic: no seed, no budget. Every median carries a 95% bootstrap interval over diagrams (1000 resamples).

Tool controls. The same graphs with the same box sizes and edge directions laid out by `neato` (`-Gstart=1`, so the file reproduces), by `neato` with `-Goverlap=false` (prism), by `dot`, and by ELK layered 0.12.0 (`elkjs`, direction RIGHT, every other option at its default), the engine the BPMN and KIELER editors run; a check in the binary confirms each control is the same graphs.

Descent, secondary. Through the library: hill climbing from the hand layout, one node per proposal, first step $d/2$ per coordinate, the step halved after 200 rejected proposals, each node kept within a disc of radius d about its start by the repair, 4000 energy evaluations, 15 seeds per instance; and uniform sampling in the same cap at the same budget on the same seeds, the matched-budget control. Reported per instance: the fraction of the cap used (mean over nodes of the distance moved over d), the fraction of energy removed, for the climber and for the control, and the number of seeds on which the climber ends lower than the control with its exact sign test. The climber pilot used a step of 0.1, five times the cap, so that nearly every proposal landed on the cap circle; the base-energy result did not depend on it (98.5 to 95.9% of the cap used as the step fell from 0.1 to 0.005) and the alignment result did (47 to 18%), which is why the directional test is primary. The converged reference is the same climber uncapped, 60,000 evaluations, with q_d measured where it stops.

Fitting. Weights on the simplex, the linear program of Section 2 over every node and direction of a corpus (about 120,000 inequalities, HiGHS), the residual its value per node; because the residual depends on the terms' scales and q does not, q under the fitted weights is reported beside it, on the fitting diagrams and held out under 5×2 cross-validation (five seeded halvings, each half fitted and scored on the other). Inverse optimisation is degenerate whenever a term is satisfied exactly: overlap holds every node, so any weight vector with all its weight on overlap has residual zero and the program returns it. The informative output is the sweep, q as the weight on one term grows from 0 with the rest on crossings and overlap, which is the test of whether any positive weight on that term can be carried.

Statistics. The unit of inference is the diagram. Per corpus, the paired comparison of q_d between the hand layout and a tool control is the Wilcoxon signed-rank test, zeros dropped, ties by midranks, exact on the observed ranks up to 50 non-zero pairs and the normal approximation with continuity and tie corrections above, with the exact sign test beside it and the Hodges–Lehmann shift with its distribution-free 95% interval; the family is the terms compared against one control within one corpus, Holm-corrected. The script carries 630 self-checks against enumeration and

against R. Equivalence claims (“no reweighting repairs it”) are read off the sweep, never off a failure to reject.

6 Measurements

The directional test’s numbers are from one deterministic run over the whole 15 to 40 node band on 2026-08-22, on chords, with the earlier BPMN corpus (147 graphs, two fifths of them EPC and Petri-net models read as bipartite graphs); the climber numbers are the pilot’s. **[PENDING: every table below is replaced from example/diagrams/results/ (the battery of run_all.sh: routes, the flow term, ELK, the 300 BPMN models, the descent with its control, the fit and its sweeps, the paired tests)]**

Each criterion alone. $q_{0.02}$ on the hand layout, with its 95% bootstrap interval over diagrams, and on the three tool layouts of the same graphs; in parentheses the median value of the term at the layout, 0 when satisfied.

corpus	energy	hand		neato	prism	dot
WikiPathways	crossings alone	1.00	[1.00, 1.00] (0.038)	1.00 (0.000)	1.00 (0.000)	1.00 (0.118)
	overlap alone	1.00	[1.00, 1.00] (0.000)	0.60 (0.030)	1.00 (0.000)	1.00 (0.000)
	length alone	0.00	[0.00, 0.00] (0.219)	0.24 (0.011)	0.19 (0.014)	0.00 (0.301)
	stress alone	0.00	[0.00, 0.00] (0.217)	1.00 (0.042)	0.94 (0.044)	0.00 (0.239)
	orthogonality alone	0.16	[0.13, 0.19] (0.193)	0.00 (0.280)	0.00 (0.281)	0.14 (0.251)
	alignment A1 alone	0.52	[0.48, 0.54] (0.003)	0.06 (0.011)	0.07 (0.010)	1.00 (0.000)
	alignment A3 alone	0.20	[0.18, 0.23] (0.284)	0.03 (0.446)	0.03 (0.451)	0.40 (0.159)
	gridiness alone	0.87	[0.84, 0.89] (0.424)	0.78 (1.000)	0.79 (1.000)	1.00 (0.059)
Reactome	node-edge alone	0.76	[0.71, 0.81] (0.022)	0.89 (0.000)	1.00 (0.000)	0.62 (0.029)
	crossings alone	1.00	[1.00, 1.00] (0.000)	1.00 (0.000)	1.00 (0.000)	1.00 (0.120)
	overlap alone	1.00	[1.00, 1.00] (0.000)	0.76 (0.022)	1.00 (0.000)	1.00 (0.000)
	length alone	0.00	[0.00, 0.00] (0.234)	0.35 (0.005)	0.26 (0.009)	0.00 (0.301)
	stress alone	0.00	[0.00, 0.00] (0.212)	1.00 (0.034)	0.85 (0.042)	0.00 (0.239)
	orthogonality alone	0.05	[0.04, 0.06] (0.225)	0.00 (0.277)	0.00 (0.277)	0.13 (0.262)
	alignment A1 alone	0.21	[0.19, 0.24] (0.004)	0.06 (0.009)	0.06 (0.010)	1.00 (0.000)
	alignment A3 alone	0.10	[0.08, 0.12] (0.300)	0.03 (0.430)	0.03 (0.428)	0.35 (0.142)
BPMN	gridiness alone	0.65	[0.62, 0.68] (0.640)	0.77 (1.000)	0.75 (1.000)	1.00 (0.062)
	node-edge alone	0.85	[0.82, 0.88] (0.006)	0.93 (0.000)	1.00 (0.000)	0.65 (0.023)
	crossings alone	1.00	[1.00, 1.00] (0.000)	1.00 (0.000)	1.00 (0.000)	1.00 (0.065)
	overlap alone	1.00	[1.00, 1.00] (0.000)	0.39 (0.082)	1.00 (0.000)	1.00 (0.000)
	length alone	0.00	[0.00, 0.00] (0.244)	0.39 (0.005)	0.27 (0.010)	0.00 (0.150)
	stress alone	0.00	[0.00, 0.00] (0.189)	1.00 (0.017)	0.86 (0.023)	0.00 (0.188)
	orthogonality alone	0.41	[0.36, 0.44] (0.140)	0.00 (0.286)	0.00 (0.286)	0.14 (0.225)
	alignment A1 alone	0.82	[0.76, 0.91] (0.001)	0.07 (0.011)	0.07 (0.011)	1.00 (0.000)
	alignment A3 alone	0.27	[0.23, 0.32] (0.214)	0.00 (0.452)	0.00 (0.450)	0.38 (0.188)
	gridiness alone	0.90	[0.85, 0.93] (0.231)	0.78 (1.000)	0.79 (1.000)	1.00 (0.111)
	node-edge alone	0.86	[0.82, 0.88] (0.006)	0.91 (0.000)	1.00 (0.000)	0.81 (0.006)

The sums. The same test under the summed energies, equal weights.

corpus	energy	hand	neato	prism	dot
WikiPathways	C+O	1.00 [1.00, 1.00]	0.60	1.00	1.00
	C+O+L	0.00 [0.00, 0.00]	0.17	0.19	0.00
	C+O+S	0.00 [0.00, 0.00]	0.60	0.94	0.00
	C+O+R	0.17 [0.14, 0.19]	0.00	0.00	0.15
	C+O+A1	0.50 [0.48, 0.53]	0.04	0.07	0.94
	C+O+A3	0.25 [0.21, 0.28]	0.03	0.03	0.53
	C+O+grid	0.84 [0.81, 0.87]	0.45	0.79	0.97
	C+O+N	0.76 [0.71, 0.80]	0.55	1.00	0.64
Reactome	C+O	1.00 [1.00, 1.00]	0.76	1.00	1.00
	C+O+L	0.00 [0.00, 0.00]	0.33	0.27	0.00
	C+O+S	0.00 [0.00, 0.00]	0.76	0.89	0.00
	C+O+R	0.06 [0.05, 0.07]	0.00	0.00	0.14
	C+O+A1	0.22 [0.20, 0.25]	0.05	0.07	0.93
	C+O+A3	0.11 [0.09, 0.12]	0.03	0.03	0.58
	C+O+grid	0.64 [0.61, 0.67]	0.52	0.75	0.97
	C+O+N	0.84 [0.81, 0.88]	0.71	1.00	0.65
BPMN	C+O	1.00 [0.94, 1.00]	0.39	1.00	1.00
	C+O+L	0.00 [0.00, 0.00]	0.23	0.27	0.00
	C+O+S	0.00 [0.00, 0.03]	0.43	0.91	0.00
	C+O+R	0.41 [0.36, 0.47]	0.00	0.00	0.14
	C+O+A1	0.75 [0.69, 0.82]	0.05	0.09	1.00
	C+O+A3	0.33 [0.29, 0.40]	0.03	0.03	0.50
	C+O+grid	0.83 [0.79, 0.88]	0.30	0.79	0.94
	C+O+N	0.81 [0.74, 0.84]	0.38	1.00	0.82

Read down a hand column. Overlap holds every box at value zero; so it does in every tool layout that removes overlaps, and **neato** without removal is the only layout it does not hold, so crossings and overlap together separate a hand layout from that configuration and from nothing else. Length and stress hold no hand-placed box, and **neato**'s stress layouts are held under stress at 1.00, so the zero is the layout's and not the test's; **dot**, which is layered and prices no distance, scores 0.00 on both as the hand layouts do. At the hand layout the three base terms share the energy as length 0.94, crossings 0.06, overlap 0.00, and $C + O + L$ inherits the length term's zero. Alignment A1 holds 0.52, 0.21 and 0.82 of hand-placed boxes, 0.06 of **neato**'s and, because a layer is a row, 1.00 of **dot**'s; the gridiness value puts 58%, 36% and 77% of hand-placed boxes in an alignment of three or more, none of **neato**'s, 94%, 94% and 89% of **dot**'s. The gridiness q of 0.77 to 0.78 on **neato**'s layouts, at value 1.000, is the step term's flatness, which is why alignment is compared on the value. Node-edge separation is nearly satisfied in every layout (0.000 to 0.029) and holds 0.62 to 1.00 for the same reason. Crossings hold every box at this radius; the value is 0.038, 0.000 and 0.000 by hand, 0.000 for **neato**, 0.118, 0.120 and 0.065 for **dot**.

Radius. q_d at $d \in \{0.005, 0.01, 0.02, 0.05\}$, hand layouts, then **neato**'s (`profile.py -sweep`):

energy	WikiPathways				Reactome				BPMN			
	.005	.01	.02	.05	.005	.01	.02	.05	.005	.01	.02	.05
overlap, hand	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
length, hand	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
stress, hand	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
alignment A1, hand	0.52	0.54	0.52	0.44	0.12	0.17	0.21	0.19	0.86	0.86	0.82	0.75
gridiness, hand	0.94	0.91	0.87	0.75	0.82	0.75	0.65	0.57	1.00	0.97	0.90	0.82
overlap, neato	0.60	0.60	0.60	0.60	0.75	0.76	0.76	0.76	0.39	0.39	0.39	0.40
length, neato	0.04	0.12	0.24	0.47	0.07	0.19	0.35	0.58	0.08	0.21	0.39	0.70
stress, neato	0.90	0.96	1.00	1.00	0.96	1.00	1.00	1.00	0.94	0.97	1.00	1.00
alignment A1, neato	0.04	0.05	0.06	0.06	0.03	0.05	0.06	0.07	0.04	0.06	0.07	0.06
gridiness, neato	0.93	0.86	0.78	0.65	0.90	0.83	0.77	0.61	0.92	0.87	0.78	0.68

The hand verdicts do not move with the radius. **neato**’s under length do: a smooth term near a minimum with gradient g and curvature λ holds a node when $|g| < d\lambda/2$, so a layout that is nearly a minimum is held at a large radius and free at a small one, and **neato**’s layouts, which minimise stress and not length uniformity, read as nearly uniform at $d = 0.05$ (0.47 to 0.70) and not at $d = 0.005$ (0.04 to 0.08). A hand layout at 0.00 across the range is far from the minimum at every scale the test resolves. Gridiness falls with d in every layout because a larger move reaches an alignment from further away.

Reference length. Under the median edge length instead of the fitted L , **neato**’s stress layouts score 0.07, 0.08 and 0.12 under stress and the hand layouts 0.00; under length the hand layouts score 0.03, 0.04 and 0.06 and **neato**’s 0.26, 0.41 and 0.43. Under $L = 1/\sqrt{N}$ no box of any layout is held under either term, the scale being wrong for every layout. The hand verdicts do not depend on the choice; the control’s does, which is why the fit is primary (Section 2).

Chance. From the climber pilot on the 10 to 400 node band: the hand layout beats uniform random placement at the same evaluation budget on 350 of 373^P pathways, and descent from the hand layout ends lower than descent from a random start on 313 of 373^P. **[PENDING: the in-cap matched-budget control through cjitter_compare on the 15 to 40 band]**

Reweighting. Held-out fraction of the cap a capped descent still uses (the secondary estimand; 0 would be stationary), from the pilot’s random-search fit on 40 diagrams per split:

energy	WikiPathways	Reactome	BPMN
asserted (1, 1, 1) ^P	98.7	91.7	92.8
base three, fitted ^P	99.0	91.7	90.4
+ node–edge separation ^P	98.0	91.3	93.0
+ orthogonality ^P	88.5	85.8	65.1
+ alignment (A1) ^P	61.0	79.9	46.9
all six ^P	57.2	79.7	58.4

Uniform edge length took weight 0.00 in every fit. The first table says why, and it also says the residual in this table is the length term’s: remove it and the base energy holds the layout outright. **[PENDING: the linear-program residual, per corpus, with and without T_L]**

7 What else could produce this

1. **The editor snapped to a grid**, so alignment is the tool’s. WikiPathways has 2% integer coordinates and a third of boxes exactly aligned, so there it is the person’s. BPMN is snapped. Test: jitter every box by half a grid cell and repeat; the alignment that survives is the person’s.
2. **The reference length.** A distance term’s verdict could be its scale’s. Tested: under the median edge length and under $1/\sqrt{N}$ the hand layouts are held under length and stress on 0.00 to 0.06 of boxes, as under the fit; only the control moves.
3. **Straight segments where the person drew polylines.** GPML and SBGN store waypoints and BPMN routes edges orthogonally; crossings, orthogonality and node–edge separation are here evaluated on centre-to-centre segments the person never saw. Test: the same table on the drawn polylines. **[PENDING:]**
4. **The person minimised a constrained energy whose constraints are absent:** reading direction, layers, compartments, lanes. The BPMN y -alignment of 0.73 is the lanes. Test:

a flow term (penalty on edges running against the reading direction) and containment as a hard constraint, added as candidates. **[PENDING:]**

5. **The radius.** At $d = 0.02$ on a thousand-pixel canvas the move is twenty pixels, about one grid step; a person cannot be stationary below the resolution they control. Test: the table at every cap (the sweep above), and at d in units of box width and of median edge length. **[PENDING:]**
6. **Selection.** Near-trees make overlap and crossings easy. Test: q_d against m/n within each corpus; and the tool controls, which are the same graphs and are held under overlap wherever the tool removes it.
7. **The tools are the wrong tools.** `dot` and `neato` are not what any of the three communities' editors run. Test: ELK layered, the BPMN editors' engine, when it is available. **[PENDING:]**

8 Related work

Aesthetic criteria and their weighting go back to Tamassia, Di Battista and Batini [3] and were made an explicit energy by Davidson and Harel [2]; Purchase [4] gave the criteria their metrics, and that list has no alignment metric. Whether people agree with the criteria has been asked three ways. By having people draw: van Ham and Rogowitz [11] found hand layouts with fewer crossings than the force-directed algorithm produced; Dwyer et al. [12] had 32 participants lay out one graph and 194 judges rank the results against three automatic layouts, and found the judges' choices track stress; Purchase, Pilcher and Plimmer [20] had participants draw from adjacency lists and found crossing removal the strongest aesthetic and alignment to a grid important; Kieffer et al. [8] collected 17 designers' layouts of 8 small graphs, ranked by 66 judges, and found their gridiness, the fraction of nodes in some alignment of three or more, correlated with rank on all 8 graphs, which is where HOLA's alignment step comes from. Alignment as a human criterion is therefore known since 2012; what is new here is the count of how many hand-placed boxes it holds, on diagrams of 15 to 40 boxes, beside the counts for the criteria tools weight. By showing people drawings: Purchase et al. [13] ran a Turing test in which hand drawings were distinguishable from algorithmic ones and judged better; Klammler, Mchedlidze and Pak [9] fitted the weights of a linear aesthetic combination by Nelder–Mead to discriminate good from worsened layouts, without human coordinates, and wrote that human-labelled data “may become a critically important accomplishment in the graph drawing field”. By learning from examples: Masui [10] learned a weighted constraint combination from user-supplied good and bad layouts by genetic programming, and Spönemann, Duderstadt and von Hanxleden [21] adjusted weights from user selections among candidates; neither tested whether the examples were minima of the learned energy. The closest prior descent from a hand layout is Flud [14]: simulated annealing continued from crowd layouts of pathway diagrams raised an asserted five-term score, on three networks, read as a success of the hybrid; the cap, the per-term test, the reference and the controls here are what turn that observation into a measurement of the score. Van Wagoningen, Mchedlidze and Telea [15] show drawings with identical metric values that differ in quality, which is why a fitted energy with no residual would still not be the energy people use, and why the claims here stay on the negative side. Kobourov, Pupyrev and Saket [22] found crossings matter little on sparse large graphs, consistent with finding 5. Inverse optimisation, the recovery of an objective from solutions taken as optimal, is due to Ahuja and Orlin [17] for linear programs and to Keshavarz, Wang and Boyd [18] for the convex case; Chan, Lee and Terekhov [19] define the goodness-of-fit residual reported here. Matched-budget random controls are standard in hyperparameter search [6] and were prescribed as the baseline for genetic algorithms in 1995 [7]; we have not found them in layout. No paper fitting energy weights to human-placed coordinates

was found in four searches of 2024 to 2026 work; the nearest, Zhang et al. [23], learns from 64,222 pairwise preference labels, not coordinates.

9 Reproducibility

Parsers, the energy (one function per term), the directional test, the descent, the fitting scripts, the parsed corpora and every table’s script are in the repository under `example/diagrams`. The directional test is deterministic; the descents are deterministic from their seed on every platform. Raw corpus archives are fetched by a script and checked by hash. **[PENDING: the commit hash of the run]**

10 Conclusion

People draw at a minimum of overlap, exactly, as every tool that removes overlaps does. They draw nowhere near a minimum of uniform edge length or of stress, the criteria that carry the energy in force-directed tools, whose own layouts the test holds. What they do that the base energy does not price is align boxes into rows and columns, for about half the boxes, by hand where the canvas is continuous; the layered tool aligns all of them. On every criterion measured here a hand layout reads as a layered layout with partial alignment. A tool builder has two changes to make and one number to report: take the weight off edge length, add alignment, and print the fraction of boxes the energy holds when started from a layout a person accepted.

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